ON THE SPACE OF METRICS WITH NON-POSITIVE CURVATURE

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Abstract. Let $\mathcal{M}(X)$ denote the space of complete Riemannian metrics with non-positive sectional curvature and with negatively curved ends, on a manifold X. We show that $\mathcal{M}(\mathbb{R}\times S^1)$ and $\mathcal{M}(\mathbb{R}\times Y)$ are path disconnected, where Y is compact and admits a negative curvature metric. The proof uses Fuller index theory as the main ingredient. Furthermore, we get a new metric deformation invariant based on geodesic string counting, and this gives a basic tool (likely to be very extendable) to further study the topology of $\mathcal{M}(X)$.

1. Introduction

There has been some study of the topology of the space of positive and non-negative curvature metrics on a given manifold X, see for instance [3], [5] which show that this topology can be very intricate. It seems that the space of non-positive curvature metrics is far less studied, in fact I don't have a single reference.

Let $\mathcal{M}(X)$ denote the space of Riemannian, complete metrics with non-positive sectional curvature and with negatively curved ends, on a manifold X. The latter means that on $X - K$, for some compact K, the sectional curvature is strictly negative. $\mathcal{M}(X)$ will be given the topology of uniform C^∞ convergence, see Definition 2.2. The latter is a metrizable topology coarser than the often used Whitney C^∞ topology. [1]

Theorem 1.1. *$\mathcal{M}(\mathbb{R}\times S^1)$ is path disconnected. Furthermore for any compact manifold Y admitting a metric of negative curvature $\mathcal{M}(\mathbb{R}\times Y)$ is path disconnected.*

Remark 1.2. We should be able to take any compact Y admitting a non-positive sectional curvature metric, our methods are almost but not quiet general enough to prove this. The issue is analogous to Euler characteristic being unable to distinguish the empty set from a torus. We need to categorify our invariant F below, and there are candidates for this, but still conjectural.

The core concept involves demonstrating the existence of a certain geodesic string counting metric deformation invariant, which uses elements of the Fuller index theory. Here a *geodesic string* in X, g is just short for: an equivalence class of a constant speed smooth g-geodesic mapping $S^1 \to X$, under the S^1 reparametrization action.

Theorem 1.3. *Let X be a smooth possibly non-compact manifold. There is functional*

$$F : \mathcal{M}(X) \times \pi_1^*(X) \to \mathbb{Q},$$

where $\pi_1^*(X)$ denotes the set of non-trivial free homotopy classes of loops in X, satisfying the following. Suppose that $\{g_t\}_{t\in[0,1]}$ is a continuous path in $\mathcal{M}(X)$, then

$$F(g_0, \beta) = F(g_1, \beta).$$

[1] We can also use uniform C^2 topology, on the other hand it appears that the weak Whitney topology does not work.

This is analogous to the invariant studied in [4], with one fundamental difference. Whereas the version of the invariant in [4] is at least partly topological and conjecturally wholly topological, in the setup here F will be shown to be decidedly non-topological. Indeed, this is crucial for the proof of Theorem 1.1.

2. The space of geodesic strings and F

Let X be a smooth manifold, which throughout is assumed to be path connected. In what follows all metrics are elements of $\mathcal{M}(X)$, with the latter as in the Introduction. Let $L_\beta X$ denote the space of all smooth maps $S^1 \to X$,

$$S^1 = \mathbb{R}/\mathbb{Z},$$

in free homotopy class β, (with usual C^∞ compact open topology). For $\beta \in \pi_1^*(X)$, $S(g,\beta) \subset L_\beta X$ will denote the subspace of all constant speed parametrized, g geodesics in class β. Set

$$\mathcal{O}(g,\beta) = S(g,\beta)/S^1,$$

the quotient by the S^1 action, where S^1 is acting by reparametrization: $t \cdot o(\tau) = o(\tau + t)$. The elements of $\mathcal{O}(g,\beta)$ are called geodesic strings.

We say that a metric g on X is β-***regular*** if all of its class β closed geodesics are non-degenerate in the usual sense, or equivalently the closed orbits of the associated geodesic flow are dynamically non-degenerate.

For a geodesic string o, $\operatorname{mult}(o)$ will denote its multiplicity, that is the order of the corresponding isotropy subgroup of S^1.

When g is β-regular, we have the following definition:

$$F(g,\beta) = \sum_{o \in \mathcal{O}(g,\beta)} \frac{(-1)^{\operatorname{morse}(o)}}{\operatorname{mult}(o)},$$

where $\operatorname{morse}(o)$ denotes the Morse index of o, (meaning the Morse-Bott index of the associated critical S^1 submanifold of $L_\beta(X)$). For this to make sense $\mathcal{O}(g,\beta)$ needs to be finite. In fact we have the following compactness theorem, which in particular implies that $\mathcal{O}(g,\beta)$ is finite for g β-regular. For a general $g \in \mathcal{M}(X)$, $F(g,\beta)$ is defined as the Fuller index, see [4, Section 5] for the definition in a totally analogous context.

Given a continuous path $\{g_t\}_{t \in [0,1]}$ in $\mathcal{M}(X)$ set:

$$S(\{g_t\},\beta) := \{(o,t) \in L_\beta X \times [0,1] \,|\, o \in S(g_t,\beta)\}.$$

Theorem 2.1. *Suppose that $\{g_t\}_{t \in [0,1]}$ is a continuous path in $\mathcal{M}(X)$. Then the space $S(\{g_t\},\beta)$ is compact.*

Proof. We need to establish uniform boundedness and the equicontinuity condition for the subspace $S(\{g_t\},\beta) \subset L_\beta(X)$. Compactness will then be immediate from the Arzella-Ascolli theorem.

Since we are dealing with constant speed parametrized g_t geodesics, (and since we are working with the compact interval $[0,1]$ for the parameter space of $\{g_t\}$), equicontinuity is readily implied by the following condition. There is a length bound, that is the length function

$$L : S(\{g_t\},\beta) \to \mathbb{R},\ (o,t) \mapsto L_{g_t}(o)$$

is bounded. This condition is verified in the first part of the proof of [4, Theorem 1.11].

Thus we just to establish a uniform point-wise bound, that is the images in X of elements of $S(\{g_t\},\beta)$ are all contained in some compact $K \subset X$.

We first show that this holds for $S(g,\beta)$ that is for a fixed metric $g \in \mathcal{M}(X)$. If there is a single class β g-geodesic string then this is obviously satisfied. Suppose there are at least two geometrically distinct class β g-geodesic strings γ_0, γ_1. Let K be a compact set s.t. g is negatively curved on $X - K$. Then by the flat strip theorem [2, Proposition 5.1] the images of γ_i must be contained in K. Since this holds for any pair this proves our claim.

We go back to the case of a family $\{g_t\}$. Suppose not, then there is a sequence $\{(o_n, t_n)\}$ in $S(\{g_t\}, \beta)$ s.t. there is no compact K with $\{o_n\}$ being contained in K. Using compactness of $[0,1]$ we may find a subsequence $\{(b_j, \tau_j)\}$, s.t. $\tau_j \mapsto t' \in [0,1]$.

First we recall the following.

Definition 2.2. *The topology of uniform C^k convergence on $\mathcal{M}(X)$ is exactly what it sounds like, but here are the details. Let $\mathrm{Sym}^2(TX)$ be the C^∞ bundle of symmetric 2-tensors. There is a natural embedding $j_k : C^\infty(X, \mathrm{Sym}^2(TX)) \to C(X, \mathrm{Jet}^k(\mathrm{Sym}^2(TX)))$, where $\mathrm{Jet}^k(\mathrm{Sym}^2(TX))$ is the k-Jet bundle. Then a neighborhood basis $\mathcal{N}_g^k$ of g is consists of sets of the form:*

$$N_{g,\epsilon} = \{h \in \mathcal{M}(X) \mid \sup_{x \in X} d_{aux}(j_k(g)(x), j_k(h)(x)) < \epsilon\},$$

for $\epsilon > 0$, and d_{aux} a metric on $\mathrm{Jet}^k(\mathrm{Sym}^2(TX))$ induced by some fixed auxiliary Riemannian metric on X. The topology of uniform C^∞ convergence is then defined by setting a neighborhood basis of g to be: $\cup_k \mathcal{N}_g^k$.

Lemma 2.3. *Set $g' = g_{t'}$. Then*

$$\inf_{o \in \beta} L_{g'}(o) > 0, \tag{2.4}$$

in particular there is a smooth closed geodesic $\gamma_1 \in \beta$.

Proof. Suppose otherwise. By continuity of $\{g_t\}$, if j is sufficiently large then $g_{\tau_j} \in N_{g',\epsilon}$ for some ϵ. Take $o \in \beta$ s.t.

$$L_{g'}(o) + \epsilon \cdot L_{g'}(o) < L_{g_{\tau_j}}(b_j),$$

which we can do by the main assumption. So that $L_{g_{\tau_j}}(o) < L_{g_{\tau_j}}(b_j)$, but this contradicts minimality of b_j in its homotopy class, which is a consequence of the Cartan-Hadamard theorem. The last part of the lemma is a consequence of the standard curve shortening argument. □

Let $\{C_n\}_{n \in \mathbb{N}}$ be an exhaustion of X by compact sets, s.t. each $C_n - C_{n-1}$ is path disconnecting, and s.t. interior $C_n \supset C_{n-1}$. For example $C_n = \{y \in X \mid d(y, x_0) \leq n\}$, for some fixed $x_0 \in X$ and some fixed Riemannian distance functional d_{aux}. Set $B_n = \overline{C_n - C_{n-1}}$.

Now, by the continuity of $\{g_t\}$ for every $\epsilon > 0$ there is a n s.t. $g_{\tau_j} \in N_{g',\epsilon}$, for all $j > n$. By the definition of $N_{g',\epsilon}$, and since the g_{τ_j} geodesic curvature of b_j vanishes, the g' geodesic curvature of b_j satisfies:

$$\lim_{j \to \infty} |\kappa_{g'}(b_j)| = 0 \text{ in } C^\infty(S^1, \mathbb{R}). \tag{2.5}$$

Lemma 2.6. *For each $k \in \mathbb{N}$, there is a $n > k$ and a $T < 0$ s.t. the sectional curvature of g' satisfies $K_{g'} < T$ on B_n.*

Proof. This is immediate by the condition that the ends of X with respect to each g_t are negatively curved. □

Now, let k be such that C_k contains the images of all elements of $S(g', \beta)$. Let n_0, T correspond to $k+1$ as in the lemma above. Fix j_0 such that the element b_{j_0} is not contained in C_{n_0}. By (2.5) we may suppose that the geodesic curvature function of $\gamma_0 = b_{j_0}$ is as small as we like.

We will need an approximate variant of the flat strip Theorem [2, Proposition 5.1].

Lemma 2.7. *Let $\alpha, \beta : S^1 \to H$ be smooth maps to a complete non-positively curved X, g, s.t.:*

(1) *α, β are homotopic.*

(2) $\max\{\mathrm{length}(\alpha), \mathrm{length}(\beta)\} < D$.

(3) *The geodesic curvatures satisfy $|\kappa_g(\alpha)| < \delta$ and $|\kappa_g(\beta)| < \delta$.*

Let $\epsilon > 0$ and $D > 0$ be given, then there is a $\delta > 0$ s.t. if α, β are as above there is an immersed cylinder $H : S^1 \times [0,1] \to X$ with boundary α, β satisfying the following. Let $v, w \in \mathrm{image}\, H_* \subset T_x X$ be a g-orthonormal pair of elements then the sectional curvature satisfies:

$$|K_g(v,w)| < \epsilon.$$

Proof. We only sketch the proof, as this is a very minor variation of [2, Proposition 5.1]. Fix a homotopy from α to β. This also determines a relative homotopy class A_t of paths between $\alpha(t)$ and $\beta(t)$. Suppose without loss of generality that $\alpha(0)$ and $\beta(0)$ are connected by a class A_0 geodesic that is a minimizer of the length function on the set:

$$\{\gamma \,|\, \gamma \text{ is a path from image}\, \alpha \text{ to image}\, \beta\}.$$

(We may adjust the parametrization and the homotopy to insure this.) For $t \in \mathbb{R}$, let $h_t : [0,1] \to X$ be the constant speed, class A_t geodesic between $\alpha(t)$ and $\beta(t)$.

Define $H(t,s) : \mathbb{R} \times [0,1] \to X$ to be the map $H(t,s) = h_t(s)$. Replacing convexity in the proof of [2, Proposition 5.1] by approximate convexity, the argument of [2, Proposition 5.1] readily implies the claim about the sectional curvature along H. ☐

Let H correspond to γ_0, γ_1 as in the lemma above. Since γ_1 is contained in C_k by assumptions, and since each $C_n - C_{n-1}$ is path disconnecting, the image of H intersects B_{n_0}. By the conclusion of Lemma 2.7 we may suppose that for a point $x \in \mathrm{image}\, H \cap B_n$, some sectional curvature of g' at x is less than $|T|$. But this contradicts the condition that $K_{g'} < T$ on B_{n_0}.

☐

Proof of Theorem 1.3. Given the compactness of Theorem 2.1, this is just the standard consequence of the invariance of the Fuller index. For details see the proof of [4, Theorem 1.14]. ☐

3. Sample calculations

Let $X = \mathbb{R} \times S^1$, and β the class of the loop $\gamma(t) = (0,t)$. Let g_0, g_1 be the familiar metrics as depicted in Figure 1. So that g_0 is negatively curved and g_1 is non-positively curved. We may also understand these metrics as the warped products $g_0 = g_{\mathbb{R}} \times_{e^t} g_{S^1}$, and $g_1 = g_{\mathbb{R}} \times_{(x^2+1)} g_{S^1}$. Here $g_{S^1}, g_{\mathbb{R}}$ are the standard metrics. The best introduction to warped products may still be the original source [1].

There are no closed β class geodesics for g_0 so

$$F(g_0, \beta) = 0.$$

g_1 has a single non-degenerate, class β geodesic string, represented by $t \mapsto (0,t)$. So

$$F(g_1, \beta) = 1.$$

Partially generalizing the above example we may take $X = \mathbb{R} \times Y$, where Y is compact and admits a negative sectional curvature metric g_{neg}. Define the warped product metric $g_0 = g_{\mathbb{R}} \times_{e^t} g_{neg}$. This metric has negative sectional curvature, for explicit computation of the sectional curvature of the warped product see [1, Lemma 7.4]. Let β be the class of $t \mapsto (0, \gamma(t))$ where γ is some simple closed g_{neg} geodesic in Y (as usual constant speed parametrized). There are no class β g_0 geodesic strings in

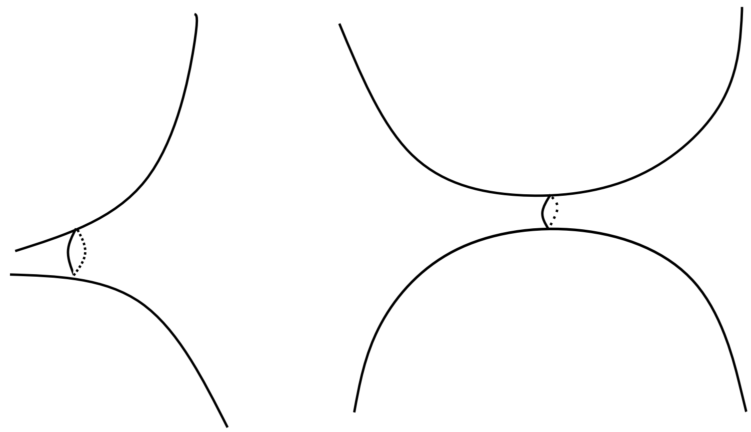

FIGURE 1. On the left is an illustration of g_0, isometrically embedded in $\mathbb{R}^3$, on the right is g_1.

X, since any such geodesic string must be minimizing by the Cartan-Hadamard theorem, and this is clearly impossible. So again $F(g_0, \beta) = 0$.

Now set $g_1 = g_{\mathbb{R}} \times_{(x^2+1)} g_{neg}$. Again by [1, Lemma 7.4] this metric has non positive sectional curvature. In this case there is a single, non-degenerate class β g_1 geodesic string represented by $t \mapsto (0, \gamma(t))$, with γ as above. So we get $F(g_1, \beta) = 1$.

4. Proof of Theorem 1.1

Let g_0, g_1, β be as in previous section. Suppose that g_0, g_1 are connected by a continuous path $\{g_t\}$ in $\mathcal{M}(X)$. Using Theorem 1.3 we get that

$$F(g_0, \beta) = F(g_1, \beta),$$

but this contradicts the calculation in the previous section. □

References

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